

Convolution to Cryptography (conv2crypt)

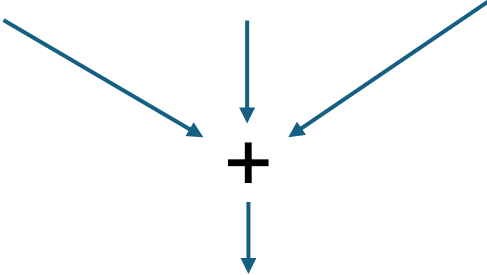
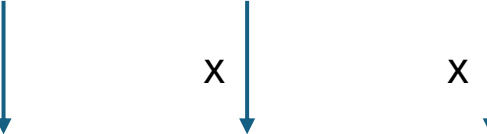
Amin Esmaeili

Matrix

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Kernel

1	1	1
---	---	---

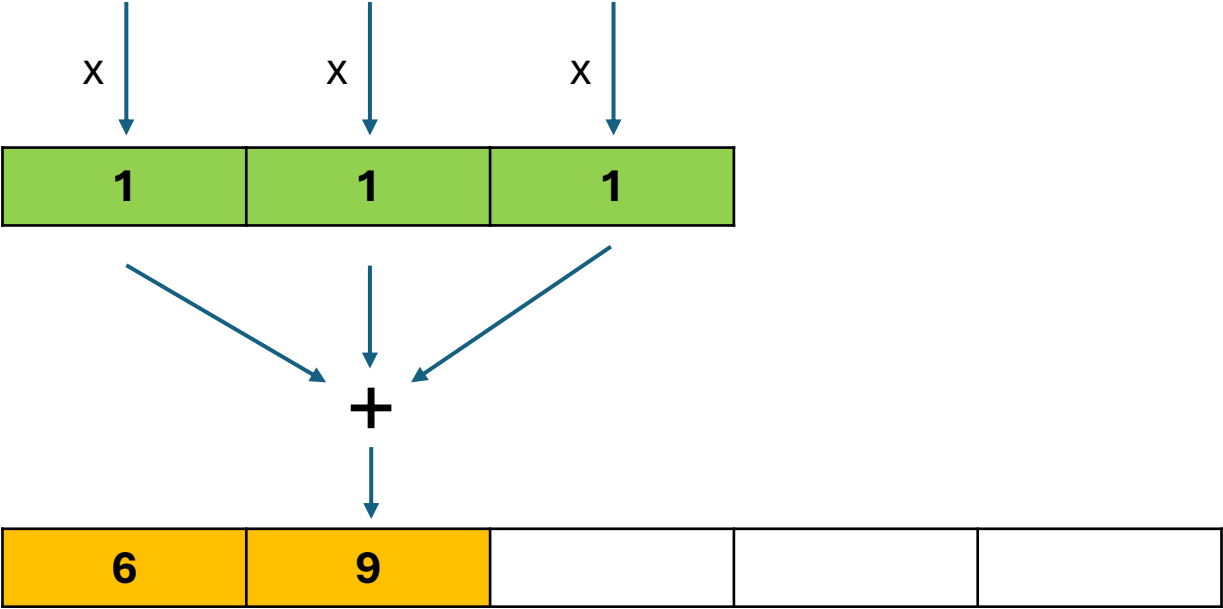


6				
---	--	--	--	--

Matrix

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Kernel



Matrix

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Kernel

1	1	1
---	---	---

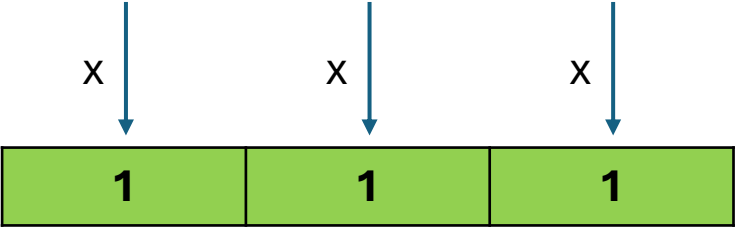
+

6	9	12		
---	---	----	--	--

Matrix

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Kernel



+

6	9	12	15	
---	---	----	----	--

Matrix

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Kernel

x	x	x
1	1	1

+

6	9	12	15	18
---	---	----	----	----

Decryption

1	2	3	4	5	6	7
---	---	---	---	---	---	---

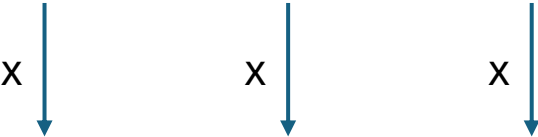
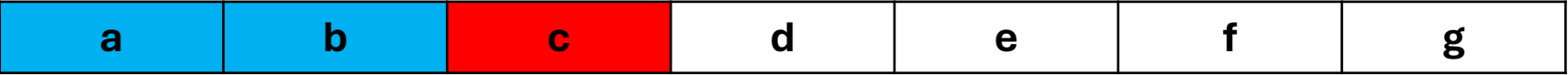
x	x	x
1	1	1

+

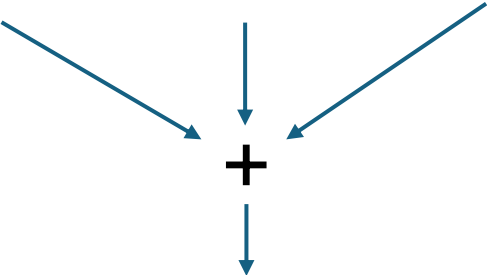
6	9	12	15	18
---	---	----	----	----

Encryption

Cipher



Kernel / Key



Plain Text



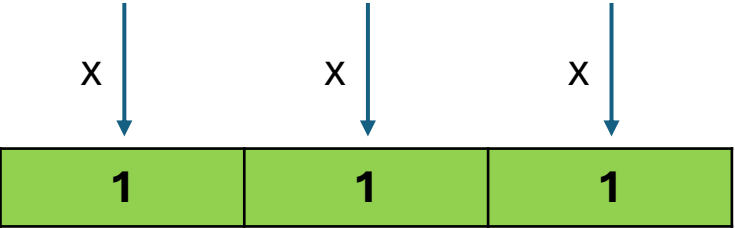
Encryption

$a = \text{random}()$
 $b = \text{random}()$
 $6 = (a \times 1) + (b \times 1) + (c \times 1) \rightarrow c = 6 - (a \times 1) - (b \times 1)$

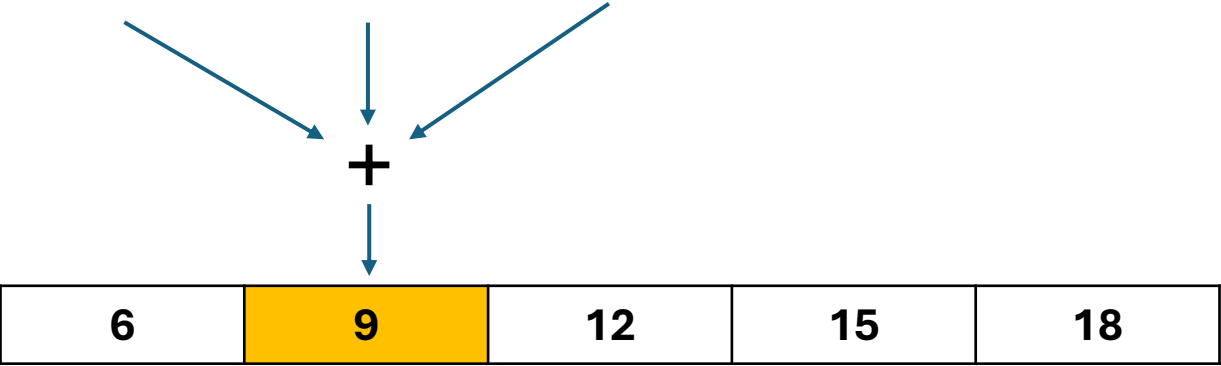
Cipher



Kernel / Key



Plain Text



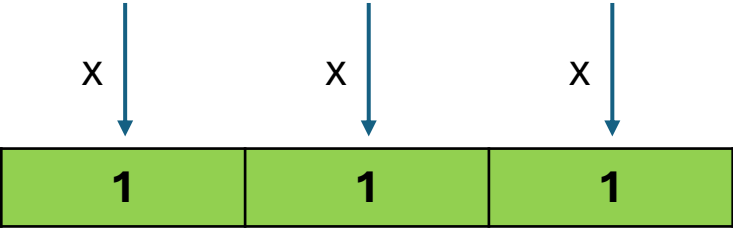
Encryption

$$b = \text{random}()$$
$$c = 6 - (a \times 1) - (b \times 1)$$
$$9 = (b \times 1) + (c \times 1) + (d \times 1) \rightarrow d = 9 - (b \times 1) - (c \times 1)$$

Cipher



Kernel / Key



+

Plain Text



Encryption

$$c = 6 - (a \times 1) - (b \times 1)$$
$$d = 9 - (b \times 1) - (c \times 1)$$
$$12 = (c \times 1) + (d \times 1) + (e \times 1) \rightarrow e = 12 - (c \times 1) - (d \times 1)$$

Cipher

a	b	c	d	e	f	g
---	---	---	---	---	---	---

Kernel / Key

1	1	1
---	---	---

Plain Text

6	9	12	15	18
---	---	----	----	----

$$d = 9 - (b \times 1) - (c \times 1)$$
$$e = 12 - (c \times 1) - (d \times 1)$$
$$15 = (d \times 1) + (e \times 1) + (f \times 1) \rightarrow f = 15 - (d \times 1) - (e \times 1)$$



Cipher

a	b	c	d	e	f	g
---	---	---	---	---	---	---

Kernel / Key

x	x	x
1	1	1

Plain Text

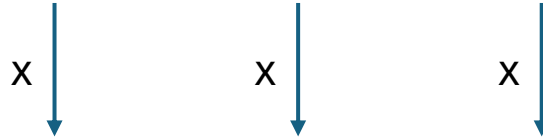
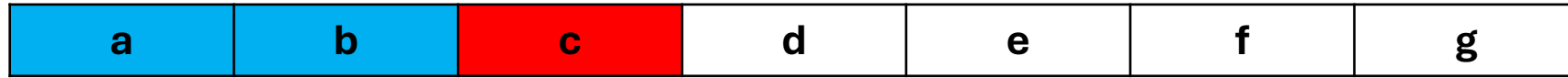
6	9	12	15	18
---	---	----	----	----

+

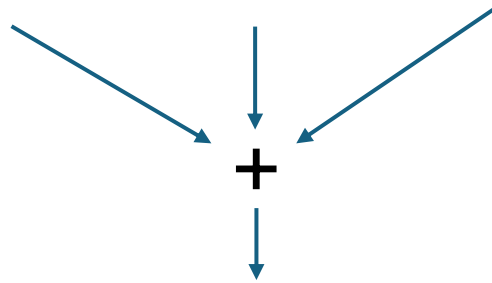
$$e = 12 - (c \times 1) - (d \times 1)$$
$$f = 15 - (d \times 1) - (e \times 1)$$
$$18 = (e \times 1) + (f \times 1) + (g \times 1) \rightarrow g = 18 - (e \times 1) - (f \times 1)$$



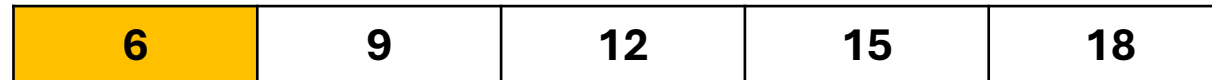
Cipher



Kernel / Key
General form



Plain Text



Encryption

$a = \text{random}()$

$b = \text{random}()$

$$6 = (a \times x) + (b \times y) + (c \times z) \rightarrow c = (6 - (a \times x) - (b \times y)) / z$$

z = 1

**** The only RULE for conv2crypt , the last index of the kernel / key must be one.
If $z = 5$, then $5 / 5 = 1$ and $6 / 5 = 1$. Then we have data loss. The only possible solution is that z be one.**